Rudra Vadhadiya

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GitHub Profile
LinkedIn Profile

Technical Skills

Programming Languages : C , C++ , Java, Javascript , Python , Flutter
Databases : MongoDB, MySQL
Web Technologies : HTML5, CSS3, React.js, Node.js, Express.js
Frameworks : Bootstrap , Tailwind CSS
DevOps : Git, GitHub
Data Science & ML : Numpy, Pandas, Matplotlib, Requests
Technical Skills : Data Structures (DS), Design and Analysis of Algorithms (DAA)

Projects

Synapse — AI Memory Layer & Context Retrieval Engine

Node.js, Express.js, PostgreSQL, pgvector, Docker, Chrome MV3, Transformers.js

- Built a privacy-first AI memory layer and context engine with a **Chrome Manifest V3** extension to capture and parse web pages using **Mozilla Readability**
- Engineered an on-device embedding pipeline with **all-MiniLM-L6-v2** generating 384-dimensional dense vectors locally without external API latency or subscriptions
- Architected a containerized **PostgreSQL + pgvector** database in Docker, indexing vectors with **IVFFlat** and cosine similarity for fast semantic retrieval
- Implemented an authenticated **Express.js REST API** with Bearer tokens and API keys, powering grounded **RAG** context retrieval with verified citations

GeoChat — Geo-Based Ephemeral Chat Application

Flutter • Dart • Firebase Firestore • Firebase Auth • Cloud Functions • BLoC

- Built a real-time, location-gated group chat app where users discover and join chat rooms within a 5 km radius using **Geohash** -based proximity queries on Cloud Firestore
- Implemented real-time message delivery via Firestore's **gRPC/WebSocket** listeners (.snapshots() stream), enabling push-based communication without polling or custom sockets
- Designed a self-destructing server architecture with 2-hour **TTL** enforced by scheduled Firebase Cloud Functions, with Firestore security rules preventing unauthorized deletions
- Architected the app using the **BLoC** state management pattern with clean separation of UI, business logic, and data layers; supported phone **OTP authentication via Firebase Auth**

Crop Disease Detection System — Smart India Hackathon (SIH)

Python, PyTorch, FastAPI, EfficientNet-B0, timm, Albumentations, OpenCV

- Developed an AI-powered REST API that detects crop diseases from leaf images with top-3 predictions and confidence scores using a fine-tuned EfficientNet-B0 model (ImageNet pretrained).
- Designed and integrated a disease-remedy knowledge base, delivering actionable treatment recommendations alongside each prediction.
- Built a FastAPI backend with Swagger UI documentation, supporting GPU-accelerated (CUDA) inference with automatic CPU fallback.
- Implemented a real-time training progress dashboard (HTML/JS) and custom data augmentation pipeline using Albumentations for improved model generalization.

Education

The Maharaja Sayajirao University of Baroda | | 2023-2027
Bachelor of Computer Science and Engineering

- GPA : 6.59

The BM Patel School | | 2023
Higher Secondary Certification (GSEB), Gujarat

- Percentile : 98.18 | ACPC Rank : 474
- GUJCET : 107.50/ 120